

End-to-end modeling of kilonova light curves and waveforms from neutron star mergers and its possible constraints on nuclear physics parameters

by

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Submitted to the Department of Physics
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ABSTRACT

Binary neutron stars mergers are interesting astrophysical events where different physical phenomena at different scales meet. General relativity with equation of state of nuclear matter dictate the hydrodynamic evolution while at the same time the neutrino transport and the nuclear statistical equilibrium establish the composition. Once the temperature has dropped the abundance and thermodynamic evolution of the ejected material will be established by the reaction network predicting the r-process nucleosynthesis and its consequence nuclear heating.

Thesis supervisor: Neelima Govind Kelkar

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Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

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Chapter 1

Introduction

In this chapter we justify our work based on current open questions. Is the abundance of heavy nuclei in the universe explained by neutron star mergers or do we need an extra astrophysical event? Is it possible, with future observations, to infer important physics results such as possible transitions to Quark-Gluon plasma?

As one may imagine, such relevant questions need accordingly rigorous answers, that of course needs to be verified with observations constantly.

Neutron star mergers are promising sites to test theoretical results from a variety of fields in physics. It has been proposed that some unclear results may have a crucial impact on the evolution of the merger. As an example, the unknown

This work attempts to contribute to the answer to such questions by reconstructing the signals from a neutron star merger for different cases. **address what do you mean by cases?**

1.1 The abundance of heavy elements in the universe

This section will present and discuss the state of the art regarding the issue with the abundance of the heavy nuclei in the cosmos.

1.2 Neutron Star Mergers as proof for fundamental physics

This section will present and discuss the state of the art of the fundamental physics results inferred by observations of neutron star mergers.

Chapter 2

Binary Neutron Stars

Since our object to study are Binary Neutron Stars mergers (BNSMs) we need to know: How frequent are these events? How do they form? How old can those objects be?

The objective of this section is to have the context of the astrophysical nature of the objects that we are studying.

2.1 The formation of the binary

This section will present the state of the art of the formation of the binary neutron stars. Also it will present how old can the binaries be.

2.2 Abundance of binaries in the universe

This section will present the latest observations and the inferred frequency of such events.

Chapter 3

Multimessengers Physics from BNSMs

Neutron star mergers are of great interest since its evolution is govern by physical phenomena from different fields. The mass ejection mechanisms is dictated by the hydrodynamics in curved spaces while the equaiton of the state of the fluid encapsules the effective interactions between its particles. The decouple and expansion of the ejected material then allows the particles inside the fuild to recombine to form proper nuclei and later atoms which impacts the evolution of the light thourgh the abundance.

In this chapter we will explain the physical mechanisms that allows the emission of the three possible messengers. Gravitational waves, Neutrinos and Electromagnetic signals

3.1 Gravitational waves

3.1.1 Detections of gravitational waves from BNSM

Update of what have been observed so far, and what we expect to observe in the future.

3.1.2 Gravitational waves mechanims

3.2 Neutrinos from BNSMs

3.2.1 Current and Future neutrino detectors

Update of what have been observed so far, and what we expect to observe in the future.

3.2.2 Possible implications from neutrino detections from BNSM

3.3 Electromagnetic signals

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Chapter 4

General Relativity Magnetic Hydrodynamics

4.1 Foundations of GRMHD

4.1.1 Equation of state for nuclear matter

4.1.2 Neutrino treatment

4.2 Numerical methods

4.3 Simulation setup

4.4 Results

Chapter 5

Nuclear Reaction Networks

5.1 Foundations of nuclear reaction networks

5.1.1 Reaction rates

5.1.2 Thermodynamic evolution

5.2 Calculation setup

5.3 Results

Chapter 6

Radiative Transfer

6.1 Foundations of radiative transfer problems

6.2 Simualiton setup

6.3 Results

Chapter 7

The end-to-end model of the merger

7.1 The merger

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.